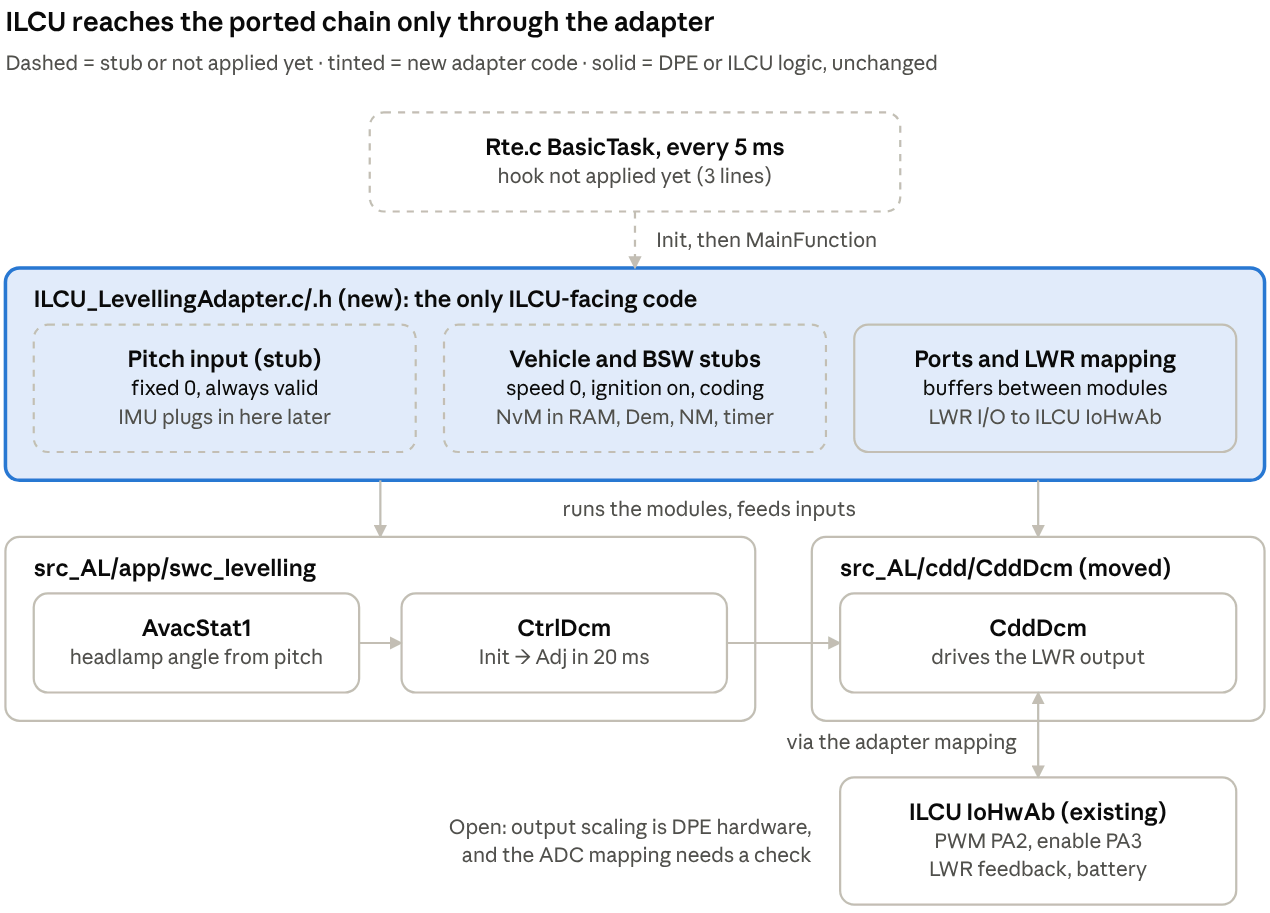
ASL Levelling Port Report (DPE → ILCU)

Nov 6, 2026 · @Sabin

The ASL chain (AvacStat1 → CtrlDcm → CddDcm) is ported into ILCU: AvacStat1, CtrlDcm and the adapter in `src_AL/app/swc_levelling`, CddDcm in `src_AL/cdd/CddDcm`. That is 27 copied files plus one adapter. Host checks pass, but the real Keil build has not run and 20 inherited DPE warnings remain by your decision.

At a glance



ported levelling chain · 4 layers, 1 integration point

The adapter runs the three DPE modules and feeds them stub inputs. CddDcm reaches the LWR hardware only through the adapter's IoHwAb mapping.

Status against the brief

Eight of ten criteria are met. The two open ones are the Keil build and the warning count, both left open by your decisions.

Criterion (from the brief)	Status	Evidence
Port only AvacStat1, CtrlDcm, CddDcm into ILCU	Met	AvacStat1, CtrlDcm and the adapter in `src_AL/app/swc_levelling/`; CddDcm in `src_AL/cdd/CddDcm/` (moved on request). Pitch and every other DPE module left out
Never modify DPE	Met	SHA-256 manifest of all 1,712 DPE files identical before and after
Modify no existing ILCU file except build config	Met	0 of 670 pre-existing ILCU files changed; the build-config lines are documented, not applied (no `Build/` tree in the workspace)
One adapter, `ILCU_LevellingAdapter.c/.h`, with pitch stub and RTE/BSW replacements	Met	Only ILCU-facing code; pitch stub returns a fixed 0
AvacStat1 reduced to Init, then Adjustment permanently	Met	Met through adapter inputs: the phase machine lives in CtrlDcm, which reaches Adjustment in 20 ms and stays there
CtrlDcm and CddDcm logic unchanged	Met	Diff against DPE: include lines and 5 renamed call sites only
Every deviation marked `/* ILCU_PORT */`	Met	Audit of all 27 copied files: 0 unmarked changes
ILCU builds with the new files	Open	Keil build not run: no `Build/` tree, no TI MCAL, Keil is Windows-only; host clang check accepted instead
No new warnings in the ported code	Open	0 in ILCU-written code; 20 inherited DPE warnings kept by decision
PORT_CHANGES.md lists every file and stub	Met	`swc_levelling/PORT_CHANGES.md`, sections 2, 4, 5 and 10

Work done

The work ran in nine steps, from reading the DPE source to moving CddDcm into the Cdd folder. Nothing in DPE or in existing ILCU files was changed.

#	Step	What was done	Output
1	Analysis	Read the full source of AvacStat1, CtrlDcm and CddDcm (about 8,000 lines), traced every external symbol in DPE, and surveyed ILCU's RTE, IoHwAb, build docs and the Al077a schematic	7 findings that changed the brief (see Open issues)
2	Plan	Wrote PORT_PLAN.md: file list per module, every dependency with its handling, the state machines and what to remove	Approved plan, stored as `swc_levelling/PORT_PLAN.md`
3	Copy	Copied 27 files: 11 AvacStat1, 10 CtrlDcm, 6 CddDcm	`AvacStat1/`, `CtrlDcm/`, `CddDcm/` (CddDcm moved in step 9)
4	Adapter	Wrote `ILCU_LevellingAdapter.c/.h`: scheduling at DPE rates (5 ms and 20 ms), the pitch stub, static buffers for the ports between modules, the LWR output mapped to ILCU IoHwAb, replacements for NvM, Dem, NM and the timer, and the DPE type subsets	The only ILCU-facing code
5	Edits to copied files	DPE-only includes replaced by the adapter header; `NvM_WriteBlock` and 4 `Pwm_*` calls renamed (they clash with ILCU names); AvacStat1's copy of the CtrlDcm phase (`SetSpeedMode`) removed; header inline functions made `static INLINE`	Every change marked `/* ILCU_PORT */`
6	Build configuration	Wrote `Make/IlcuLevelling_030_Flags.gmk` (include paths) and `Make/IlcuLevelling_050_Files.gmk` (13 sources)	CustVar lines documented, not applied
7	Verification	Host compile check, native link at -O0 and -O2, a behaviour harness with 4 scenarios, DPE and ILCU checksum manifests	See Verification

#	Step	What was done	Output
8	Documentation	Wrote PORT_CHANGES.md: files added, edits, stubs, open issues, evidence and corrections	`swc_levelling/PORT_CHANGES.md`
9	Move CddDcm to the Cdd folder	Moved CddDcm into ILCU's CDD layout at `src_AL/cdd/CddDcm/` (`Api` , `Gen` , `Cfg` , `Make`), split the make fragments (the app pair now lists 11 sources), updated PORT_PLAN.md and PORT_CHANGES.md, and re-ran every check	Files byte-identical, no source line changed; identical warnings and harness transcripts

Decisions

You made four decisions during planning, reconfirmed two when the open criteria came up again, and later amended the folder choice.

Decision point	Your choice	Effect
Where to reduce the Init → Adjustment phase machine (it lives in CtrlDcm, not AvacStat1)	Force it through adapter inputs; CtrlDcm untouched	CtrlDcm reaches Adjustment in 20 ms and stays there; Fail only on real faults (pitch invalid, LWR output shorted). AvacStat1 lost only its copy of the phase
Target folder	`src_AL/app/swc_levelling` , one folder; later amended: CddDcm moves to `src_AL/cdd/CddDcm`	AvacStat1, CtrlDcm, the adapter and the docs stay in the app folder; CddDcm sits beside ILCU's other CDD, CddAsm330lhh
Inherited DPE warnings vs "no new warnings"	Leave DPE code as-is (reconfirmed)	20 warnings stay and are documented; the offered fix set was declined
How to verify the build	Snapshot only: host checks here, Keil build by you	Host clang check plus native harness done; Keil build pending

Decision point	Your choice	Effect
Build criterion, raised again	Accept the host check in place of the Keil build	The build criterion stays formally open

Verification

All host checks pass: 0 compile errors, 0 warnings in ILCU-written code, and 4 of 4 harness scenarios. None of them replaces the Keil build.

Check	Result	Limits
Compile check, clang `-Wall`, Cortex-M0+ target, armclang-like flags, all 13 sources	0 errors; 0 warnings in the adapter and on `ILCU_PORT` lines; 20 warnings in unchanged DPE lines	Not armclang; stand-ins for the missing TI MCAL headers
Same check with `-Wextra`	1 extra warning, also in DPE code (`-Wmissing-field-initializers`)	Informational
Native link at -O0 and -O2	Links after the `static INLINE` fix; before it, -O0 failed with undefined `Cod_AvacStat1_*`	Host linker, not armlink
Harness: healthy inputs, pitch 200 then 400 (0.1 mrad)	Adjustment at 20 ms, held for 45 s; target −115 then −229 (0.01°) after the 10 s hold; LWR move completes in 14 s	IoHwAb simulated
Harness: pitch becomes invalid	Fail, then the coded safety position −32 (0.01°)	The recovery cycle differs between -O0 and -O2 (DPE undefined behaviour)
Harness: LWR output shorted to ground	Short detected in about 0.5 s; Fail at 2.56 s	Simulated fault
Harness: ILCU output-stage model	Commanded 430 ‰ of battery voltage, reached 225 ‰ with PWM at 1000 ‰	Confirms the hardware mismatch

Check	Result	Limits
Re-run after the CddDcm move	Moved files byte-identical; 0 errors and the identical 20-warning set; 4 of 4 scenarios PASS, with transcripts byte-identical to before the move at -O0 and -O2	Same host setup as above
Checksum manifests	DPE: 1,712 of 1,712 files identical. ILCU: 0 of 670 existing files changed; 35 new files, 27 in `app/swc_levelling` and 8 in `cdd/CddDcm`	Covers every file in both trees

Not real yet

Every vehicle input and BSW service the chain needs is a stub in the adapter, and ILCU does not call the chain yet.

Item	Now	Made real by
Pitch angle (`ILCU_LevellingAdapter_GetPitchAngle`)	Fixed 0, always valid	The IMU module: define `ILCU_LVLADP_EXTERNAL_PITCH_SOURCE` and implement the function. `swc_imu` already computes `Pitch_cdeg` (× 1.745 gives 0.1 mrad; sign to verify)
Vehicle speed and dynamics	0 km/h, valid	A CAN speed signal; until then only standstill levelling runs
Ignition and vehicle network	Always on, powertrain active	A real ignition state; until then the "ignition needed" latch clears only at reboot
Coding block	DPE defaults, except the standstill speed thresholds set to 2 and 5 km/h (DPE defaults of 0 block levelling)	ILCU coding or NvM
NvM, 5 blocks	RAM only: writes return OK, nothing is stored	ILCU NvM blocks (needs an ILCU NvM config change)
Dem	Last status per event kept in RAM for debugging	An ILCU Dem
Pitch plausibility, right-side ECU signals, UDS routine control, NM, timer	Fixed healthy values	Only if the demo needs them

Item	Now	Made real by
Build wiring	Lines for `CustVar_030_Flags.gmk` and `CustVar_050_Files.gmk` documented, not applied. They pull in two fragment pairs: `app/swc_levelling/Make/IlcuLevelling_*` and `cdd/CddDcm/Make/CddDcm_*`	Add them in the real `Build/` tree, both pairs together
Scheduling hook	Not in `Rte.c`, so the linker drops the chain	3 lines in `src_AL/rte/Rte.c`: the include, `ILCU_LevellingAdapter_Init()` in `Rte_Start()`, `ILCU_LevellingAdapter_MainFunction()` after `IoHwAb_MainFunction()`

The full list of 15 stubs is in PORT_CHANGES.md, section 5.

Open issues

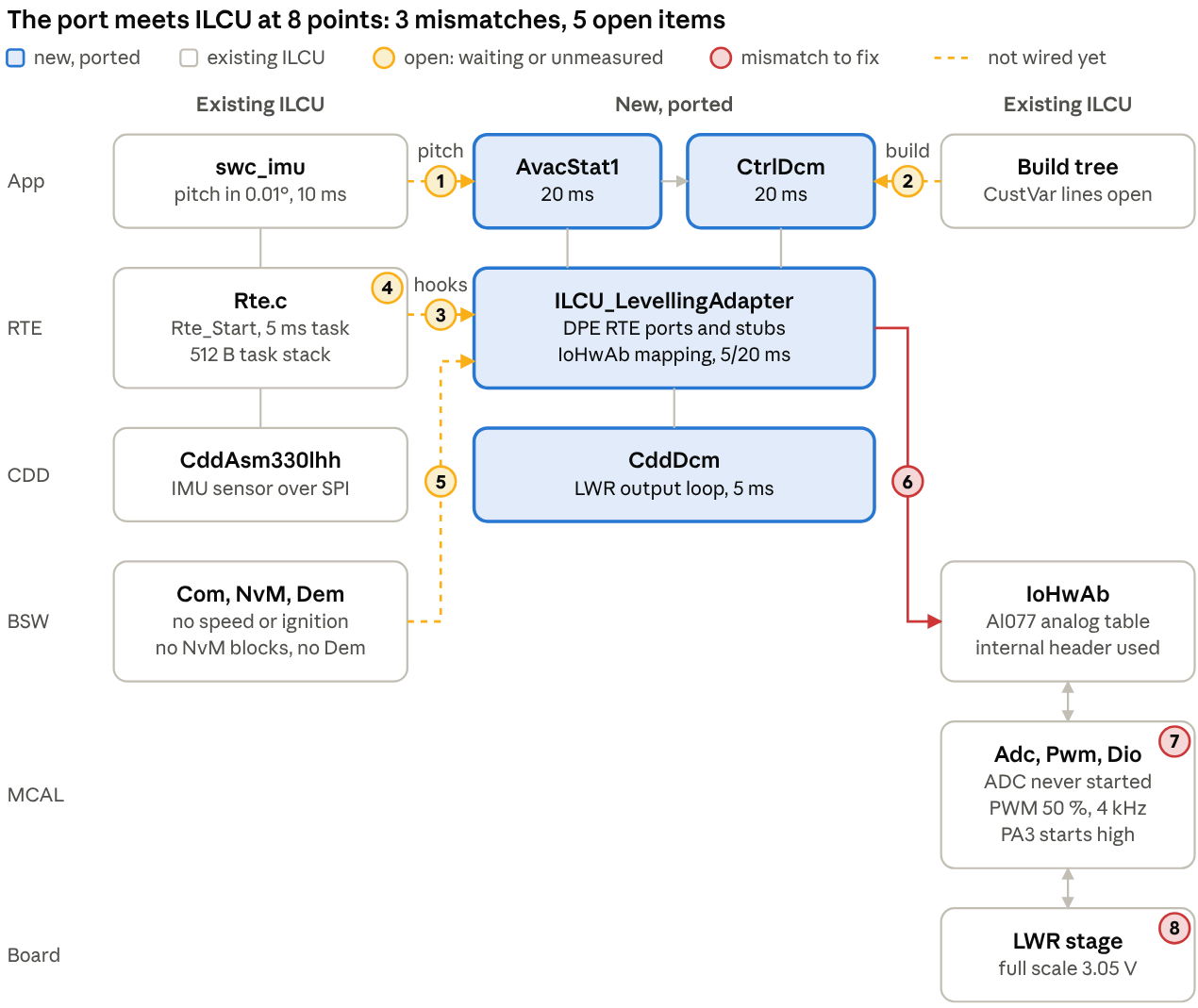
The biggest open issue is hardware: on ILCU's output stage the chain drives the lamp to the wrong position without flagging an error.

Issue	Impact	Next step
CddDcm's output stage is DPE hardware: it regulates a ratio of battery voltage, while ILCU's stage tops out near 3.05 V	Lamp ends up in the wrong position, no error raised: 225 ‰ reached for 430 ‰ commanded	Hardware or product decision before a real actuator is connected; CddDcm logic is frozen by the brief
Al077a IoHwAb analog config disagrees with the schematic: battery on ADC0 ch4 vs ch5, feedback divider 82.5k/21k vs 82.5k/15k	A battery reading outside 8.59–17.69 V locks the LWR output permanently	Check on hardware, then fix the ILCU IoHwAb config
Undefined behaviour in `CtrlDcm_HandleFailState` (`CtrlDcmSm.c:902`): uninitialized locals on fault recovery	For one 20 ms cycle, -O0 stops the output while -O2 drives it at 206 ‰	Optional fix: initialize the two locals to `CtrlDcm_eFreeze` (declined so far)
20 inherited DPE compiler warnings: unused code and uninitialized reads	The "no new warnings" criterion stays open	Kept by decision; verbatim list in PORT_CHANGES.md, section 6.4

Issue	Impact	Next step
Keil build not run	The "ILCU builds" criterion stays open	Apply the CustVar lines, run `Build.bat … Al077 App default NonKey rebuild`, check the log for `swc_levelling`
DPE defects carried over unchanged	The adjustment counter never increments; one CddDcm error flag is not cleared in one branch	Report to the DPE owners; not fixed here by rule

ILCU interface map

The ported chain meets existing ILCU at 8 points. Five are still open, among them the IMU pitch that AvacStat1 waits for. The other three are mismatches below the adapter, in the IoHwAb mapping, the MCAL settings and the output stage.



ILCU interface map · 8 points where the port meets existing ILCU

Amber points wait on another step or team; red points are configuration or hardware mismatches. Each point is explained below.

#	Kind	Where	What is open or wrong	Closed by
1	Waiting	swc_imu → AvacStat1, through `ILCU_LevellingAdapter_GetPitchAngle`	The pitch is a stub at 0. swc_imu computes `Pitch_cdeg` from the accelerometer every 10 ms (`swc_imu.c:131`) but only sends it to Com (`swc_imu.c:149` , `Rte.c:137-154`). AvacStat1 needs 0.1 mrad, front up positive. An accelerometer pitch also contains the road gradient, which DPE's axle-height pitch did not	The IMU module implements the function and the build defines `ILCU_LVLADP_EXTERNAL_PITCH_SOURCE` (`ILCU_LevellingAdapter.h:36-43`), with unit (×1.745), sign, validity and a gradient strategy
2	Waiting	Build tree → port folders	The CustVar lines for both fragment pairs are not applied	Build guide steps 2 and 3

#	Kind	Where	What is open or wrong	Closed by
3	Waiting	Rte.c → adapter	`Rte_Start` and `TASK(BasicTask)` do not call the adapter, so the linker drops the chain	Build guide step 5
4	Unmeasured	BasicTask stack	512 B (`Os_Wrapper_Cfg.c:124`); the chain alone needs 128 B, before IoHwAb, MCAL and 64 B of frames	Stack measurement, build guide step 7
5	Waiting	Com, NvM, Dem → adapter	Speed, ignition, vehicle mode and mounting side are constants; NvM is RAM only; ILCU has no Dem (stubs S3–S13)	CAN signals in Com, levelling NvM blocks, a Dem decision
6	Mismatch	Adapter → IoHwAb	The AI077 analog table reads KL15 from an unwritten slot (0 V) and scales the feedback ×4.97 instead of ×6.5; the adapter uses the internal header	Change list I2–I4 and A1–A3
7	Mismatch	IoHwAb → MCAL	ADC groups are never started; the LWR PWM is variable-period, starts at 50 % and runs at 4 kHz; PA3 starts high	Change list I1 and M1–M4
8	Mismatch	MCAL → board	The LWR stage gives D × 3.05 V where CddDcm needs about D × 17.3 V	Change list H1 (R8004)

Commit strategy

Fourteen commits on one feature branch, in this order. Start with commits 1 and 2: they change nothing in the build, and they make every later deviation from DPE reviewable as a plain diff. Commits 1–5 only add files the build does not see, commit 6 is the first to change the build, and commit 13 is the first that runs the chain.

#	Commit subject	Files	Build effect	Status
1	`docs(levelling): add DPE ASL port plan and change log`	2	None	To do
2	`chore(levelling): import DPE ECO ASL sources unmodified`	27	None, not in the build	To do
3	`feat(levelling): add ILCU_LevellingAdapter`	2	None	To do
4	`feat(levelling): adapt the DPE sources to ILCU (ILCU_PORT)`	24	None	To do

#	Commit subject	Files	Build effect	Status
5	`build(levelling): add make fragments for swc_levelling and CddDcm`	4	None	To do
6	`build(al077a): compile the levelling chain in the App variant`	2	13 objects compile; the linker drops them	To do
7	`fix(al077a/mcal): safe defaults for the LWR output`	2	LWR output off at boot, PWM at 20 kHz	To do
8	`fix(iohwab): start both ADC groups on every IoHwAb cycle`	1	Analog values start updating	To do
9	`fix(al077a/iohwab): map the ILCU analog inputs`	2	KL15 and LWR feedback read correctly	To do
10	`refactor(iohwab): public LWR and KL15 API, used by the adapter`	6	None at run time	To do
11	`fix(levelling): park the LWR output and gate MainFunction on Init`	2	Defensive start-up	To do
12	`fix(levelling): initialise CtrlDcm locals on the recovery path`	3	Removes undefined behaviour	To do
13	`feat(rte): run the levelling chain from the RTE`	1	The chain runs	To do
14	`feat(imu): feed the vehicle pitch to the levelling adapter`	2	Real pitch replaces the stub	To do

- **Every commit leaves `rebuild` green:** no `error` in the log and a fresh hex. The Check line of each commit below says what else to look at.
- **Owners.** Commits 7–9 belong to the ILCU BSW owner and can be prepared in parallel with 1–6, but must land before 13. Commit 12 needs your approval, because it reverses decision Q3. Commit 14 waits for the IMU pitch. H1 (R8004) is a hardware change, not a commit, and it gates connecting an actuator.

- **Merge without squashing.** Commits 2 and 4 together are the audit trail of every ILCU_PORT deviation, and a squash merge folds them into one diff. If squash merges are mandatory, give commit 2 its own pull request.
- **Make commits 2 and 4 with the same line-ending settings** (`core.autocrlf`). Otherwise the commit-4 diff shows every line as changed.

Phase A: add files the build does not see (commits 1–5)

1 · Port plan and change log. Files: `Src/src_AL/app/swc_levelling/PORT_PLAN.md`, `Src/src_AL/app/swc_levelling/PORT_CHANGES.md`. Before committing, correct PORT_CHANGES.md: the project folder (`Al077a`, §2 and §8) and F7 (§6.1). I can make both corrections on your OK.

```
docs(levelling): add DPE ASL port plan and change log

PORT_PLAN.md holds the approved scope of the port of AvacStat1,
CtrlDcm and CddDcm from DPE ECO and how each DPE dependency is
handled. PORT_CHANGES.md lists every added file, every stub (S1-S15),
the known issues and the host verification. No code yet.
```

2 · DPE sources, unmodified. 27 files, committed with the exact DPE bytes:

- `Src/src_AL/app/swc_levelling/AvacStat1/`: `AvacStat1.c`, `AvacStat1.h`, `AvacStat1Data.c`, `AvacStat1Data.h`, `AvacStat1Init.c`, `AvacStat1Init.h`, `AvacStat1Types.h`, `AvacStat1Util.c`, `AvacStat1Util.h`, `ctaaAvacStat1.c`, `ctaaAvacStat1.h`
- `Src/src_AL/app/swc_levelling/CtrlDcm/`: `Api/CtrlDcm.h`, `CtrlDcm.c`, `CtrlDcmSm.c`, `CtrlDcmSm.h`, `CtrlDcm_Data.c`, `CtrlDcm_Data.h`, `CtrlDcm_RteAccess.c`, `CtrlDcm_RteAccess.h`, `ctaaCtrlDcm.c`, `ctaaCtrlDcm.h`
- `Src/src_AL/cdd/CddDcm/`: `Api/CddDcm.h`, `Gen/CddDcmCtrl.c`, `Gen/CddDcmCtrl.h`, `Gen/CddDcmHal.c`, `Gen/CddDcmHal.h`, `Cfg/CddDcm_Cfg.h`

Your working tree already holds the ported versions. This script swaps in the DPE originals, commits them with the message below, and puts the ported versions back. Run it in Git Bash from the repo root, after commit 1, with the DPE `Src/Src_AL` folder as its argument. Tested on a throwaway repository: 27 files committed byte-identical to DPE, and the ported versions restored.

```
#!/bin/bash
# Commit 2: commit the DPE originals unmodified at their ILCU paths, then put
the
# ported versions back so commits 3-5 can add them. Run from the ILCU repo
root,
# after commit 1. Usage: commit2_import_dpe.sh <DPE Src/Src_AL folder>
set -euo pipefail
DPE=${1:?usage: $0 <DPE Src/Src_AL folder>}
L=Src/src_AL/app/swc_levelling
C=Src/src_AL/cdd/CddDcm
A=Cust/St/St126/App/Swc
BK=$(mktemp -d)

while read -r src dst; do
  mkdir -p "$BK/$(dirname "$dst")" "$(dirname "$dst")"
  if [ -f "$dst" ]; then cp -p "$dst" "$BK/$dst"; fi
  cp -p "$DPE/$src" "$dst"
  git add "$dst"
done <<EOF
$A/AfsMaster/AvacStat1/AvacStat1.c        $L/AvacStat1/AvacStat1.c
$A/AfsMaster/AvacStat1/AvacStat1.h        $L/AvacStat1/AvacStat1.h
$A/AfsMaster/AvacStat1/AvacStat1Data.c    $L/AvacStat1/AvacStat1Data.c
$A/AfsMaster/AvacStat1/AvacStat1Data.h    $L/AvacStat1/AvacStat1Data.h
$A/AfsMaster/AvacStat1/AvacStat1Init.c    $L/AvacStat1/AvacStat1Init.c
$A/AfsMaster/AvacStat1/AvacStat1Init.h    $L/AvacStat1/AvacStat1Init.h
$A/AfsMaster/AvacStat1/AvacStat1Types.h   $L/AvacStat1/AvacStat1Types.h
$A/AfsMaster/AvacStat1/AvacStat1Util.c    $L/AvacStat1/AvacStat1Util.c
$A/AfsMaster/AvacStat1/AvacStat1Util.h    $L/AvacStat1/AvacStat1Util.h
$A/AfsMaster/AvacStat1/ctaaAvacStat1.c    $L/AvacStat1/ctaaAvacStat1.c
$A/AfsMaster/AvacStat1/ctaaAvacStat1.h    $L/AvacStat1/ctaaAvacStat1.h
$A/Ctrl/Dcm/Api/CtrlDcm.h                 $L/CtrlDcm/Api/CtrlDcm.h
$A/Ctrl/Dcm/CtrlDcm.c                     $L/CtrlDcm/CtrlDcm.c
$A/Ctrl/Dcm/CtrlDcmSm.c                   $L/CtrlDcm/CtrlDcmSm.c
$A/Ctrl/Dcm/CtrlDcmSm.h                   $L/CtrlDcm/CtrlDcmSm.h
$A/Ctrl/Dcm/CtrlDcm_Data.c                $L/CtrlDcm/CtrlDcm_Data.c
$A/Ctrl/Dcm/CtrlDcm_Data.h                $L/CtrlDcm/CtrlDcm_Data.h
$A/Ctrl/Dcm/CtrlDcm_RteAccess.c           $L/CtrlDcm/CtrlDcm_RteAccess.c
$A/Ctrl/Dcm/CtrlDcm_RteAccess.h           $L/CtrlDcm/CtrlDcm_RteAccess.h
$A/Ctrl/Dcm/ctaaCtrlDcm.c                 $L/CtrlDcm/ctaaCtrlDcm.c
$A/Ctrl/Dcm/ctaaCtrlDcm.h                 $L/CtrlDcm/ctaaCtrlDcm.h
Cdd/CddDcm/Gen/CddDcm.h                   $C/Api/CddDcm.h
Cdd/CddDcm/Gen/CddDcmCtrl.c               $C/Gen/CddDcmCtrl.c
Cdd/CddDcm/Gen/CddDcmCtrl.h               $C/Gen/CddDcmCtrl.h
```

```
Cdd/CddDcm/Gen/CddDcmHal.c                $C/Gen/CddDcmHal.c
Cdd/CddDcm/Gen/CddDcmHal.h                $C/Gen/CddDcmHal.h
Cust/St/St126/Cdd/CddDcm/CddDcm_Cfg.h     $C/Cfg/CddDcm_Cfg.h
EOF

git commit -q -F - <<'MSG'
chore(levelling): import DPE ECO ASL sources unmodified

Byte-identical copies of the DPE ECO sources of AvacStat1, CtrlDcm
and CddDcm, placed at their ILCU paths. Nothing builds them yet. The
next commits apply the ILCU_PORT edits, so every deviation from DPE
shows as a plain diff against this commit.

Source: DPE Src/Src_AL, St126 ECO variant (27 files)
MSG

cp -pR "$BK/." .
echo "commit 2 done; ported versions restored from $BK"
```

3 · Adapter. Files: `Src/src_AL/app/swc_levelling/ILCU_LevellingAdapter.h`, `Src/src_AL/app/swc_levelling/ILCU_LevellingAdapter.c`.

```
feat(levelling): add ILCU_LevellingAdapter

The only integration point between ILCU and the ported chain:
- schedules the DPE runnables: CddDcm every 5 ms, AvacStat1 and
  CtrlDcm every 20 ms
- implements the DPE RTE ports as buffers or documented stubs
- maps the LWR output stage onto IoHwAb
- provides ILCU_LevellingAdapter_GetPitchAngle, a stub returning 0
  until the IMU module supplies the pitch

Nothing builds it yet.
```

4 · ILCU_PORT edits. 24 files: everything in commit 2 except `AvacStat1/AvacStat1.h`, `AvacStat1/ctaaAvacStat1.h` and `Cfg/CddDcm_Cfg.h`, which are unchanged from DPE. Stage them with `git add -u Src/src_AL/app/swc_levelling Src/src_AL/cdd/CddDcm`. Check: `git show --stat HEAD` lists 24 files.

```
feat(levelling): adapt the DPE sources to ILCU (ILCU_PORT)

Every change carries an ILCU_PORT comment:
- DPE-only includes replaced by ILCU_LevellingAdapter.h, include
  paths flattened
- NvM_WriteBlock and Pwm_* calls go through the adapter, because the
  names collide with ILCU NvM and the TI MCAL
- AvacStat1 reduced to Init plus permanent Adjustment: SpeedMode and
  the LvlPhase mirror removed
- AvacStat1Data.h accessors made static INLINE, so -O0 builds link

CtrlDcm and CddDcm logic is unchanged. Nothing builds it yet.
```

5 · Make fragments. Files: `Src/src_AL/app/swc_levelling/Make/IlcuLevelling_030_Flags.gmk`, `Src/src_AL/app/swc_levelling/Make/IlcuLevelling_050_Files.gmk`, `Src/src_AL/cdd/CddDcm/Make/CddDcm_030_Flags.gmk`, `Src/src_AL/cdd/CddDcm/Make/CddDcm_050_Files.gmk`. Check: `git status` shows nothing left in either port folder (35 files tracked).

```
build(levelling): add make fragments for swc_levelling and CddDcm

IlcuLevelling_030/050 list the adapter, AvacStat1 and CtrlDcm (11
sources). CddDcm_030/050 list CddDcm (2 sources) with the include
paths Api, Gen and Cfg, following the CddAsm330lhh fragments.
Nothing includes them yet.
```

Phase B: compile (commit 6)

6 · Build wiring. Files: `Build/Make/Cust/Al/Al077a/App/CustVar_030_Flags.gmk`, `Build/Make/Cust/Al/Al077a/App/CustVar_050_Files.gmk`. The folder is UNVERIFIED; use the App variant your ILCU build uses. The lines are in build guide steps 2 and 3. Check: after a `rebuild`, `_object_files.txt` lists 13 new objects and only the 20 DPE warnings appear.

```
build(al077a): compile the levelling chain in the App variant

CustVar_030_Flags.gmk and CustVar_050_Files.gmk include both fragment
pairs. The 13 sources compile, but nothing calls the chain yet, so
armlink --remove drops it from the image. Expect the 20 warnings from
unchanged DPE lines listed in PORT_CHANGES.md section 6.4.
```

Phase C: ILCU BSW and configuration (commits 7–9, BSW owner)

7 · Safe LWR output defaults. Files: `Src/config_AL/Al/Al077a/mcal/Port_Cfg/src/Port_PBcfg.c` (PA3 level, line 683), `Src/config_AL/Al/Al077a/mcal/Pwm_Cfg/src/Pwm_PBcfg.c` (lines 110, 111 and 114). Check: scope PA3 and PA2 from power-on; both stay low until CddDcm drives them.

```
fix(al077a/mcal): safe defaults for the LWR output

- EN_LWR (PA3) starts low, so the output stage stays disabled until
  software enables it (schematic section 7)
- the LWR PWM starts at 0 % instead of 50 %
- the LWR PWM period is 2000 ticks: 20 kHz instead of 4 kHz
- the LWR PWM channel is fixed-period, so a new duty cycle brings it
  back from Pwm_SetOutputToIdle
```

8 · ADC sampling. File: `Src/src_AL/ecu9_core_ptf/Src/Src_AL/bsw/IoHwAb/internal/IoHwAb_Analog.c` (platform code). If the platform cannot change, use `Src/src_AL/rte/Rte.c` instead: two `Adc_StartGroupConversion` calls before `IoHwAb_MainFunction()`, with the subject `fix(rte): start the ADC groups every 5 ms`. Check: `IoHwAb_AdcBuffer0[0]` and `IoHwAb_AdcBuffer1[0]` follow the inputs.

```
fix(iohwab): start both ADC groups on every IoHwAb cycle

Both ADC groups are software-triggered one-shot and nothing started
them, so every IoHwAb analog value stayed 0. IoHwAb now starts each
group after copying its last result.
```

9 · Analog table. Files: `Src/config_AL/Al/Al077a/bsw/IoHwAb/IoHwAb_AnalogCfg.c`, `Src/config_AL/Al/Al077a/bsw/IoHwAb/IoHwAb_AnalogCfg.h`. Keep the channel IDs `IOHWAB_CHID_VBAT` and `IOHWAB_CHID_ADIN_3` in this commit, so the adapter and

`IoHwAb_Cfg.c` still build; commit 10 renames them. Check: bench checks 2 and 3 of the change list.

```
fix(al077a/iohwab): map the ILCU analog inputs

Replaces the Al077 board table with the four ILCU inputs, in the slot
order of Adc_PBcfg.c: KL15 (ADC0 ch5, x6.5), PCB NTC (ADC0 ch6, raw),
LWR feedback (ADC1 ch0, x6.5) and 5V0 (ADC1 ch2, x2). KL15 was read
from slot 4 of a 2-channel group, so it read 0 V; the LWR feedback
used the 82.5k/21k slope.
```

Phase D: port follow-ups (commits 10–12)

10 · Public IoHwAb API and its only caller. Files: `Src/config_AL/Al/Al077a/bsw/IoHwAb/IoHwAb_AnalogCfg.h`, `Src/config_AL/Al/Al077a/bsw/IoHwAb/IoHwAb_Cfg.c`, `Src/config_AL/Al/Al077a/bsw/IoHwAb/IoHwAb_Cfg.h`, `Src/src_AL/app/swc_levelling/ILCU_LevellingAdapter.c`, `Src/src_AL/app/swc_levelling/Make/IlcuLevelling_030_Flags.gmk`, `Src/src_AL/app/swc_levelling/PORT_CHANGES.md`. The API change and its only caller land together, so every commit builds. Check: no file under `swc_levelling` includes `IoHwAb_Internals.h`.

```
refactor(iohwab): public LWR and KL15 API, used by the adapter

Renames IOHWAB_CHID_VBAT to IOHWAB_CHID_KL15 and IOHWAB_CHID_ADIN_3
to IOHWAB_CHID_LWR_FB. Adds IoHwAb_VoltageKL15_Read,
IoHwAb_VoltageLWR_Read, IoHwAb_PwmCTRL_LWR_SetDutyCycle and
IoHwAb_PwmCTRL_LWR_SetOutputToIdle, and removes
IoHwAb_VoltageKL30_Read (ILCU has no KL30). The adapter, the only
caller, switches in the same commit and no longer includes
IoHwAb_Internals.h, so the IoHwAb/internal include path goes too.
```

11 · Park the output, gate MainFunction. Files: `Src/src_AL/app/swc_levelling/ILCU_LevellingAdapter.c`, `Src/src_AL/app/swc_levelling/PORT_CHANGES.md`. Check: rebuild.

```
fix(levelling): park the LWR output and gate MainFunction on Init

ILCU_LevellingAdapter_Init now drives EN_LWR low and parks the PWM
before the CDD init, because CddDcm writes the enable only when its
state changes. MainFunction returns until Init has run, so a change
in the start-up order cannot run the chain uninitialised.
```

12 · CtrlDcm locals (needs your approval). Files: `Src/src_AL/app/swc_levelling/CtrlDcm/CtrlDcmSm.c`, `Src/src_AL/app/swc_levelling/CtrlDcm/CtrlDcm_RteAccess.c`, `Src/src_AL/app/swc_levelling/PORT_CHANGES.md`. Check: the host harness gives the same output at `-O0` and `-O2`.

```
fix(levelling): initialise CtrlDcm locals on the recovery path

eFailStateLeft, eFailStateRight and eFailState in CtrlDcmSm.c and
three locals in CtrlDcm_RteAccess.c could be read uninitialised. On
the recovery path the host harness gave different outputs at -O0 and
-O2. Each now starts at its safe value. This reverses decision Q3
for these lines only.
```

Phase E: run (commit 13)

13 · RTE hooks. File: `Src/src_AL/rte/Rte.c`, as in build guide step 5. Check: build guide steps 6 and 7, then on-target checks 1–8 with the actuator disconnected. Connect an actuator only after H1.

```
feat(rte): run the levelling chain from the RTE

Rte_Start calls ILCU_LevellingAdapter_Init first, and BasicTask calls
ILCU_LevellingAdapter_MainFunction every 5 ms, right after
IoHwAb_MainFunction. From here on the chain runs on the target.
```

Phase F: IMU pitch (commit 14, IMU team)

14 · Real pitch. Files: `Src/src_AL/app/swc_imu/swc_imu.c`, and `Build/Make/Cust/Al/Al077a/App/CustVar_030_Flags.gmk` for `C_DEFINES += ILCU_LVLADP_EXTERNAL_PITCH_SOURCE`. Before committing, agree how the road gradient is removed (interface point 1). Check: tilt the board front up and watch `ILCU_LvlAdp_LastPitch` go positive, at about 1.745 × `Pitch_cdeg`.

```
feat(imu): feed the vehicle pitch to the levelling adapter

swc_imu implements ILCU_LevellingAdapter_GetPitchAngle, and the build
defines ILCU_LVLADP_EXTERNAL_PITCH_SOURCE, which removes the adapter
stub. Pitch_cdeg is converted to 0.1 mrad (x 349 / 200), front up
positive, and reported invalid until the IMU is initialised.
```